MILITARY SPECIFICATION SHEET

CABLE, ELECTRICAL, 600 VOLTS, TYPES 2SJ, 3SJ and 4SJ

This specification sheet is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for procuring the cable described herein shall consist of this document and the latest issue of Specification MIL-C-915.

REQUIREMENTS:

Qualification required.

Construction

First - Copper conductors, tin-coated for ASTM, sizes 14 and smaller. Uncoated for Navy sizes 6 and larger.
Second - Polyvinyl chloride insulation 105°C., (see details for dimensions). Standard Identification Code applied by method 3.
Third - Two, three or four conductors, as required, shall be cabled together with fibrous fillers as necessary to form a firm, round assembly. (see details for maximum length of lay).
(B) Fourth - Braided shield of tin coated copper strands AWG No. 30, or smaller, providing a minimum coverage of 85 percent.
(B) Fifth - Binder tape of Polyester, Type G of MIL-I-631, applied helically with overlap.
Sixth - Standard thermoplastic jacket with cable surface marking.

Details

Type and size 1/	Conductor No.	Conductor Size ASTM B-286	Conductor Size Navy Std.	Dia. over insul. nom. (inch)	Lay of condr. max. (inches)	Overall diameter Min. (inch)	Overall diameter Max. (inch)	Condr. res. per 1000 ft. at 25° C. max. (ohms)
2SJ-22	2	22-19		0.067	1-1/2	0.187	0.203	16.54
2SJ-20	2	20-19		.073	1-1/2	.204	.220	10.16
2SJ-18	2	18-19		.084	1-1/2	.227	.246	6.47
2SJ-16	2	16-19		.091	2-1/2	.245	.265	5.02
2SJ-14	2	14-19		.105	2-1/2	.279	.302	3.17
2SJ-12	2		6(7)	.145	3	.380	.400	1.72
2SJ-11	2		9(7)	.160	3	.420	.445	1.25
2SJ-9	2		14(7)	.200	4	.505	.530	.79
2SJ-7	2		23(7)	.235	4	.580	.605	.50
(B) 3SJ-9	3		14(7)	.200	4	.580	.600	.79
3SJ-22	3	22-19		.067	1-1/2	.201	.217	16.54
3SJ-20	3	20-19		.073	1-1/2	.216	.234	10.16
3SJ-18	3	18-19		.084	1-1/2	.244	.264	6.47
3SJ-16	3	16-19		.091	2-1/2	.265	.285	5.02
3SJ-14	3	14-19		.105	2-1/2	.301	.325	3.17
3SJ-12	3		6(7)	.145	3	.415	.440	1.72
4SJ-20	4	20-19		.073	1-1/2	.260	.280	10.16
4SJ-16	4	16-19		.091	2-1/2	.305	.325	5.02
4SJ-14	4	14-19		.105	3	.350	.370	3.17

1/ Size designation does not correlate to conductor size for all types and sizes.

(B) Denotes changes

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EXAMINATION AND TESTS:

	Requirements:
Basic electrical:	
Conductor resistance - At 25°C., ohms/1000 ft., maximum	See details
Voltage withstand - Volts, rms, minimum,	
Conductor to conductor	1200
Conductor to shield	600
Shield to water	600
Insulation resistance - Megohms/1000 ft., minimum	500
Group A:	
Visual and dimensional	
Group B:	
Physicals (unaged)	
Insulation (conductor)	
Tensile strength - psi, minimum	1800
Elongation - percent, minimum	100
Jacket (cable)	
Tensile strength - psi, minimum	1800
Elongation - percent, minimum	100
Group C:	
Physicals (aged)	
Insulation	
Air oven	
Elongation - percent of unaged, minimum	90
Jacket	
Hot oil immersion	
Elongation, percent, minimum	50
Permanence of printing (jacket) - Cycles, minimum	250

QUALIFICATION INSPECTION

Qualification inspection shall include basic electrical plus all of Groups A, B and C.

UNIT ORDERING LENGTHS

All sizes: 1000 Feet (nominal)

ustodians:
Army - MI
Navy - SH
Air Force - 80

Review activities:
Army - EL, MI, AV
Navy - EC
Air Force - 80

User activities:
Army - ME, MU, WC
Navy - CG

Preparing activity:
Navy - SH
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